Threat Classification & Geospatial Intelligence Dashboard

Milestone 2 Report | Power BI Security Incident Analytics Extension

1. Objective

The objective of Milestone 2 was to extend the baseline incident-reporting dashboard built in Milestone 1 into a full Threat Intelligence solution. Rather than simply logging incidents, the enhanced dashboard is designed to help security leadership answer three operational questions in real time: how severe are current threats, where are they concentrated geographically, and how efficiently is the response team resolving them. To achieve this, the project layers Threat Classification (severity scoring, category-wise escalation behaviour, and critical-incident tracking) on top of Geospatial Threat Intelligence (city- and state-level risk mapping, response-time benchmarking, and hotspot identification). The result is an interactive, drill-down-capable Power BI report that supports faster triage decisions, better allocation of security resources, and clearer visibility into which regions and threat categories are driving organizational risk.

Key Performance Indicators at a Glance

AVG. CITY RISK SCORE	CRITICAL INCIDENTS	ESCALATION RATE %	AVG. RESOLUTION TIME
153.40	267	30.47%	13.38 hrs

2. Dashboard Screenshot

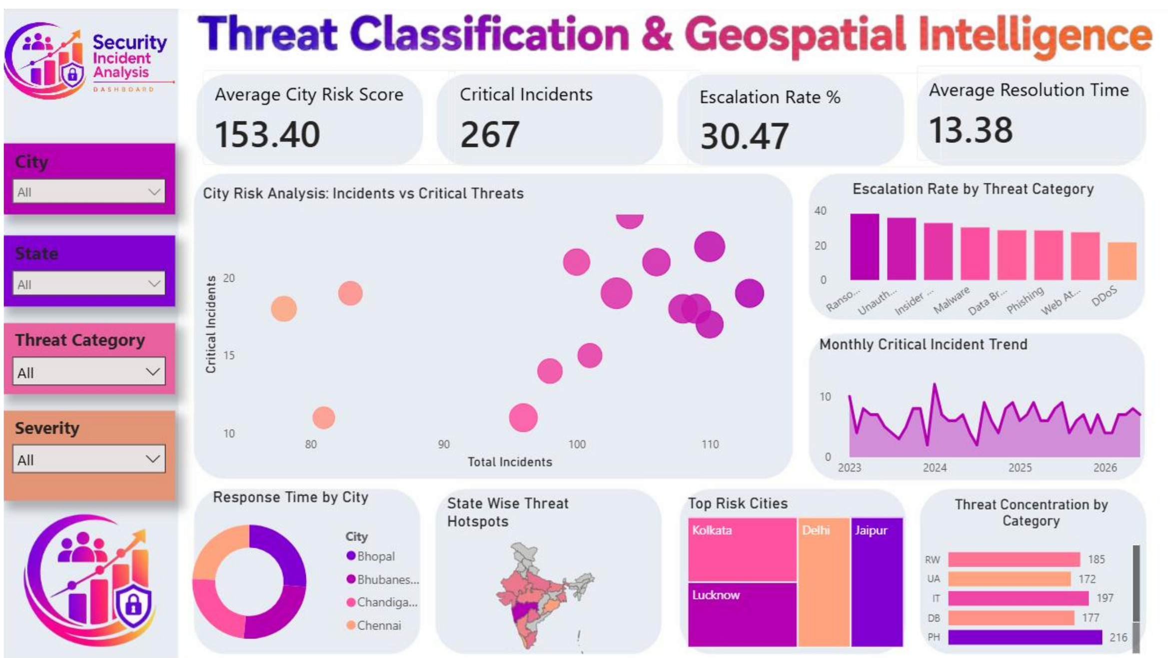

Figure 1: Threat Classification & Geospatial Intelligence Dashboard — Power BI

The dashboard is organized around four interactive filters (City, State, Threat Category, and Severity) on the left rail, allowing analysts to slice every visual simultaneously. The top KPI strip surfaces the four headline metrics, while the central canvas combines a City Risk Analysis scatter plot (Total Incidents vs. Critical Incidents), an Escalation Rate by Threat Category bar chart, and a Monthly Critical Incident Trend line chart spanning 2023–2026. The lower band adds geospatial and comparative context through a Response Time by City donut chart, a State-Wise Threat Hotspot map of India, a Top Risk Cities ranking, and a Threat Concentration by Category breakdown.

3. Three Key Insights

- **Insight 1 — Escalation is broad-based, not isolated to one category.** The Escalation Rate by Threat Category chart shows that Ransomware, Unauthorized Access, and Insider Threats all carry escalation rates in a similarly high range, closely followed by Malware, Data Breach, Phishing, and Web Attacks. This indicates the organization's escalation problem is systemic across nearly all threat types rather than confined to a single vector, with an overall escalation rate of 30.47% meaning roughly one in three incidents is being pushed beyond first-line handling.
- **Insight 2 — A distinct cluster of high-volume, high-severity cities exists.** The City Risk Analysis scatter plot reveals a clear cluster of cities with both high total incident counts (100+) and elevated critical-incident counts (20+), positioned in the upper-right region of the chart. These cities represent compounding risk — they are hit more often and more severely — and align with the cities named in the Top Risk Cities panel, namely Kolkata, Delhi, Jaipur, and Lucknow, against an average city risk score of 153.40 across the network.
- **Insight 3 — Resolution speed is lagging behind incident severity and varies sharply by city.** With an average resolution time of 13.38 hours against 267 critical incidents and a 30.47% escalation rate, response throughput is not keeping pace with the severity mix. The Response Time by City donut further shows meaningful variation between cities such as Bhopal, Bhubaneswar, Chandigarh, and Chennai, suggesting the delay is not uniform but concentrated in specific locations that would benefit from targeted process or staffing intervention.

4. Two Business Recommendations

1. **Prioritize security investment and staffing in the identified high-risk cities.** Kolkata, Delhi, Jaipur, and Lucknow consistently surface as top-risk locations across both the scatter plot and the Top Risk Cities ranking. Redirecting SOC analysts, faster escalation paths, and on-ground incident-response resources toward these cities first would address the largest share of critical incidents with the smallest operational footprint, rather than spreading resources evenly across all locations.

2. **Standardize and accelerate response protocols for the highest-escalation threat categories.** Since Ransomware, Unauthorized Access, and Insider Threats show the steepest escalation behaviour, the organization should introduce category-specific runbooks and pre-approved escalation thresholds for these threat types. Combined with city-level performance benchmarking from the Response Time by City view, this would help close the gap between the current 13.38-hour average resolution time and a tighter, more consistent target across all regions — directly reducing the 30.47% escalation rate over time.